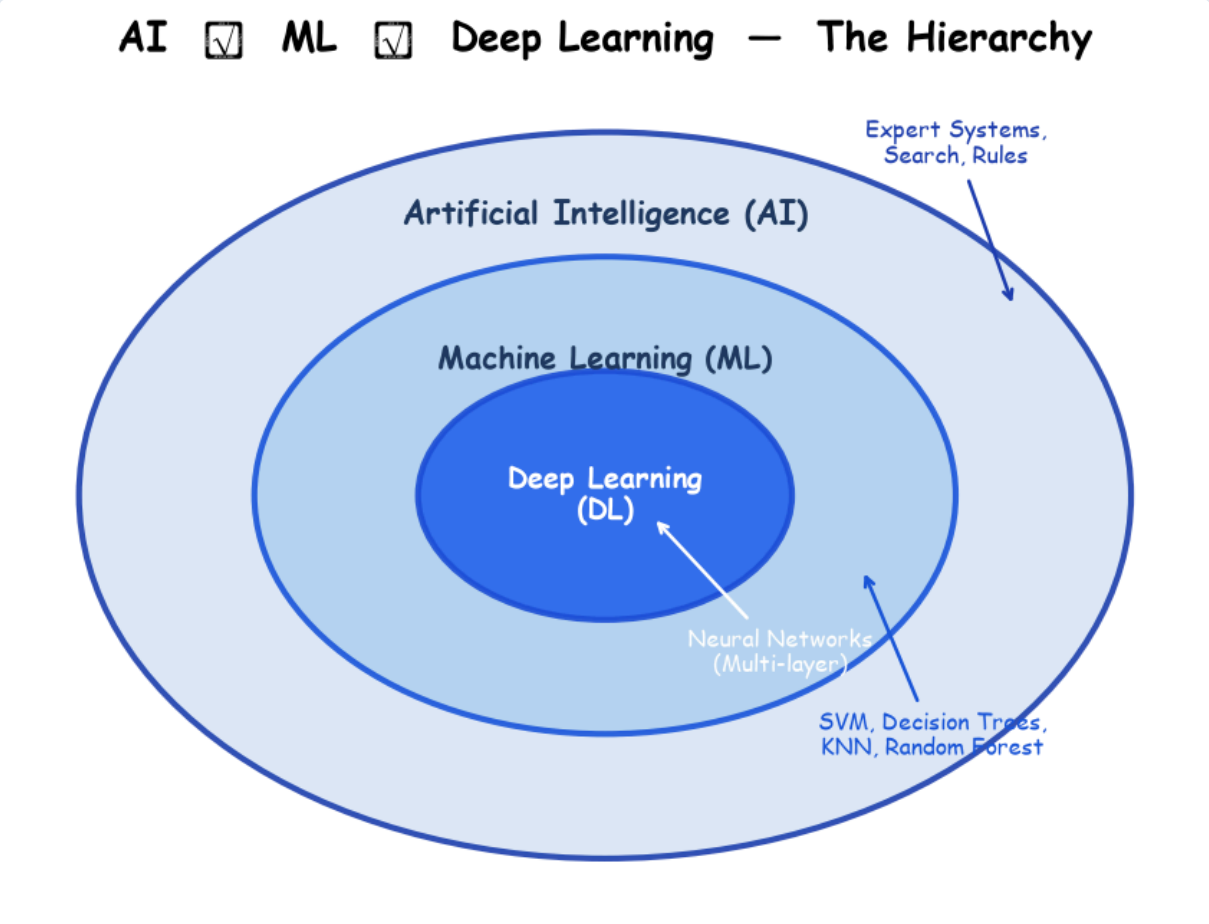
Lecture 36: Deep Learning Foundations & Artificial Neurons

Welcome to the definitive master curriculum repository for **Lecture 36: Deep Learning Foundations (Part-1)**. This lecture marks the monumental transition from classical statistical Machine Learning into modern Deep Learning and Artificial Neural Networks.

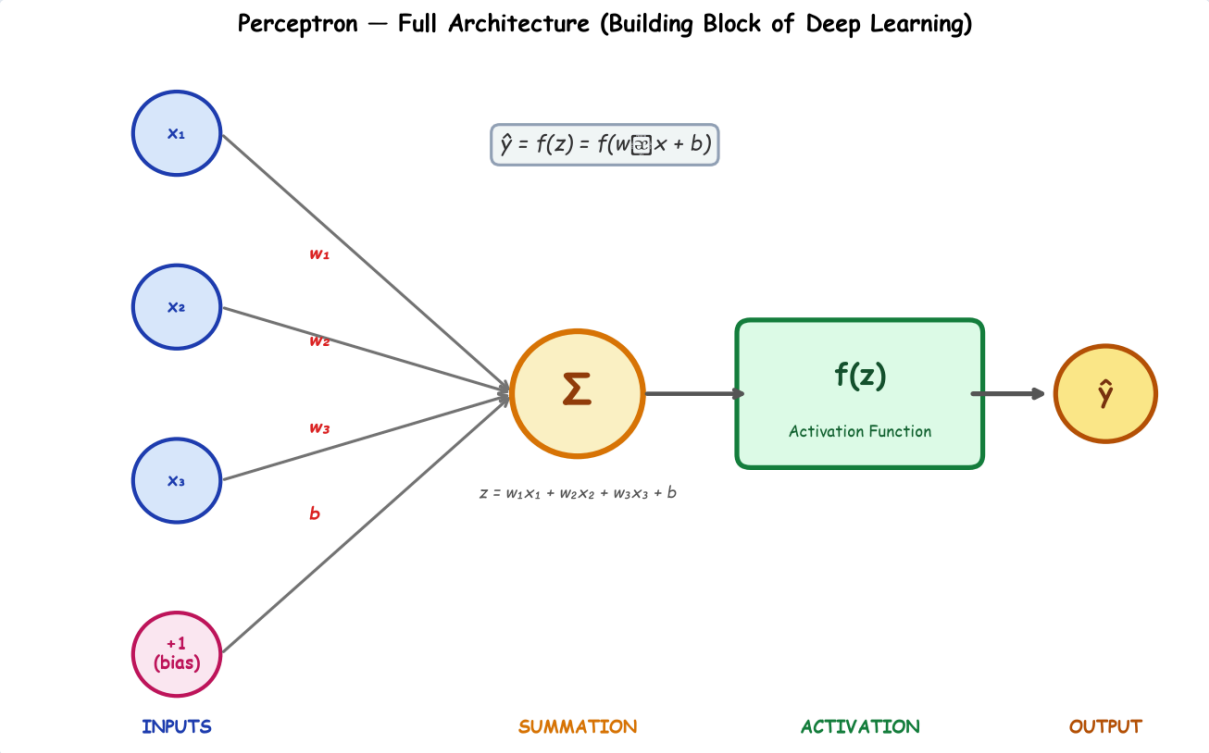
Every groundbreaking frontier in modern artificial intelligence—from autonomous driving perception systems to Large Language Models (LLMs) like GPT-4 and Claude—is powered by multi-layered neural architectures. This course builds rigorous mathematical foundations in perceptron dynamics, activation functions, non-linear representation learning, hardware acceleration, and PyTorch implementations.

Architectural Taxonomy: AI vs ML vs DL



Core Concept: The Artificial Neuron (Perceptron)

The foundational building block of all neural networks is the artificial neuron, which computes a linear combination of input features weighted by synaptic parameters, offsets the sum with a bias, and evaluates an activation function:



The Primary Mathematical Formulation:

$$z = \sum_{i=1}^{n} w_i x_i + b = \mathbf{w}^T \mathbf{x} + b$$

$$\hat{y} = f(z) = f(\mathbf{w}^T \mathbf{x} + b)$$

Comprehensive Submodule Roadmap

This curriculum is organized into 10 structured topic submodules accompanied by an industrial practice suite. Every module features dual-language documentation (English `README.md` and Hinglish `README_HINGLISH.md`), Mermaid whiteboard diagrams, executable PyTorch notebooks, and compiled vector-grade PDFs:

Submodule	Topic Title	Core Concepts & Formulas Covered	Asset Reference	Documentation Links
36.01	**What is Deep Learning?**	AI vs ML vs DL hierarchy, Big Data, GPU parallelization, real-world apps	Diagram / Concepts	English | Hinglish
36.02	**ML vs Deep Learning**	Manual feature engineering vs automated learning, Andrew Ng scaling curve	Diagram / Concepts	English | Hinglish
36.03	**What are Neural Networks?**	Biological neuron mapping, 3-layer ANN architecture (Input, Hidden, Output)	Diagram / Concepts	English | Hinglish
36.04	**Perceptron (In Depth)**	Frank Rosenblatt model, weights, bias necessity, Step, Sigmoid, ReLU, training loop	Diagram / Concepts	English | Hinglish
36.05	**Multi-Layered Networks**	XOR non-linearity limit, Multi-Layer Perceptron (MLP), Universal Approximation	Diagram / Concepts	English | Hinglish
36.06	**PyTorch vs TF vs Keras**	Dynamic vs static computation graphs, research vs enterprise production	Diagram / Concepts	English | Hinglish

Submodule	Topic Title	Core Concepts & Formulas Covered	Asset Reference	Documentation Links
36.07	**PyTorch Installation**	CPU vs CUDA GPU installation, hardware detection script	`pytorch_installation.png`	English | Hinglish
36.08	**Google Colab Setup**	Free NVIDIA T4 GPU cloud runtime, Drive integration, session limits	Diagram / Concepts	English | Hinglish
36.09	**Building a Neuron (Code)**	`torch.tensor`, `nn.Linear(3, 1)`, forward pass, mathematical proof	Notebook	English | Hinglish
36.10	**Neuron Code Walkthrough**	Cell-by-cell notebook execution, `@` matrix dot product verification	Notebook	English | Hinglish
Practice	**Industrial Scenarios**	Cardiovascular diagnostic neuron, XOR mathematical proof, GPU VRAM sizing	Multiple	English Questions | Hinglish Questions | Solutions

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Comparison: Machine Learning vs Deep Learning

Parameter	Traditional Machine Learning	Deep Learning
Dataset Scale	Effective on small to medium structured tabular data	Requires large datasets to avoid overfitting

Parameter	Traditional Machine Learning	Deep Learning
Feature Extraction	Manual, handcrafted by domain experts	Automated hierarchical feature learning directly from raw data
Hardware	Optimized for standard CPUs	Requires GPU/TPU matrix acceleration
Model Interpretability	Glass Box (clear feature weights, decision trees)	Black Box (opaque parameter matrices)
Data Modalities	Tabular, numerical, categorical data	High-dimensional unstructured data (vision, audio, NLP)